# Course: MTH101

## Short Notes For Midterm

## Lecture # 01

**What is calculus?**
Calculus is the study of the continuous rates of change of quantities.

**What does calculus study?**
Calculus studies how various quantities change with respect to other quantities.

**What are natural numbers?**
Natural numbers are the simplest numbers, such as 1, 2, 3, 4, 5, which are used for counting.

**What is a subset?**
A subset is a part of a collection or set.

**What are integers?**
Integers are numbers that include the natural numbers, their negatives, and zero.

**What are rational numbers?**
Rational numbers are numbers formed by taking ratios of integers, excluding division by zero.

**What are irrational numbers?**
Irrational numbers are numbers that cannot be expressed as the ratio of integers, such as $\sqrt{2}$ or $\pi$.

**What is analytic geometry?**
Analytic geometry is a branch of mathematics that describes algebraic formulas using geometric curves and vice versa.

**How are real numbers represented on a coordinate line?**
Each real number corresponds to a point on a coordinate line.

**What does the order of real numbers represent?**
The order of real numbers represents the relative size of one number compared to another.

**What does the symbol ">" mean?**
The symbol ">" means "greater than."

**What does the symbol "<=" mean?**
The symbol "<=" means "less than or equal to."

**What does the symbol "∪" represent?**
The symbol "∪" represents the union of two sets, combining all their elements.

**What does the symbol "∩" represent?**
The symbol "∩" represents the intersection of two sets, including only the common elements.

# Lecture # 02

**What is the notation used for absolute value?**
Answer: The notation used for absolute value is $|a|$.

**How is the absolute value of a non-negative real number defined?**
Answer: The absolute value of a non-negative real number is the number itself.

**What is the absolute value of a negative real number?**
Answer: The absolute value of a negative real number is the number without the minus sign.

**Is 0 considered a positive or negative number in mathematics?**
Answer: 0 is considered neither positive nor negative in mathematics. It is a non-negative number.

**What is the effect of taking the absolute value of a number?**
Answer: The effect of taking the absolute value of a number is to remove the minus sign if the number is negative.

**Can both +a and -a be positive or negative?**
Answer: No, if a is negative, then -a is positive and +a is negative.

**Solve the equation $|x-3| = 4$.**
Answer: The solutions to the equation $|x-3| = 4$ are $x = 7$ and $x = -1$.

**What is the relationship between square roots and absolute values?**
Answer: Every positive real number has two real square roots, one positive and one negative.

**What is the positive square root of 9?**
Answer: The positive square root of 9 is 3.

**Is the equality 2a = a always correct?**
Answer: No, the equality 2a = a is correct only when a is non-negative.

**What is the theorem related to the square of a number and its square roots?**
Answer: The theorem states that for any real number a, a^2 = (+a)^2 = (-a)^2.

**What are the properties of absolute value?**
Answer: (a) |-a| = |a|, (b) |ab| = |a||b|, (c) |a/b| = |a|/|b|.

**What is the distance formula for points A and B on a coordinate line?**
Answer: The distance formula is d = |b - a|.

**How can inequalities of the form |x-a| < k be solved?**
Answer: Inequalities of this form can be solved by finding the interval notation for the solutions.

**Solve the inequality |x-3| < 4.**
Answer: The solution to the inequality |x-3| < 4 is (-1, 7).

**Solve the inequality x + |a| ≥ 4.**
Answer: The solution to the inequality x + |a| ≥ 4 is x ≤ -a or x ≥ 6.

**Is it always true that |a + b| ≤ |a| + |b|?**
Answer: Yes, it is always true that |a + b| ≤ |a| + |b|. This is known as the triangle inequality.

**What is the Triangle Inequality theorem?**
Answer: The Triangle Inequality theorem states that for any real numbers a and b, |a + b| ≤ |a| + |b|.

**What is the essence of the Heisenberg Uncertainty Principle in quantum physics?**
Answer: The essence of the Heisenberg Uncertainty Principle is based on the Triangle Inequality.

**What is the proof of the Triangle Inequality?**
Answer: The Triangle Inequality can be proved by adding two inequalities and analyzing the cases where a + b is greater than or equal to 0 or less than 0.

# Lecture # 03

**What is the coordinate plane?**
The intersection of two coordinate lines at 90 degrees.

**How can points in the plane be represented?**
By pairs of real numbers.

**What is the purpose of drawing two lines through a point in the coordinate plane?**
To associate a unique ordered pair of real numbers with that point.

**What is the rectangular coordinate system?**
The coordinate plane along with the ordered pairs, together known as the rectangular coordinate system.

**What are quadrants?**
The four regions into which the plane is divided by the coordinate axes.

**How are graphs of equations in two variables defined?**
They are the set of all points in the plane whose coordinates satisfy the equation.
**What is the importance of intercepts in a graph?**
Intercepts are points where the graph intersects the coordinate axes.
**What are x-intercepts and y-intercepts?**
X-intercepts are points of the form (a, 0), and y-intercepts are points of the form (0, b).

**What is symmetry in the context of coordinate planes?**
Symmetry refers to the property of points being symmetric about an axis or the origin.

**How can symmetry be used as a tool for graphing?**
By taking advantage of existing symmetries, the work required to obtain a graph can be reduced.

**Is the graph of an equation always an exact representation?**
No, when obtained by plotting points, there is no guarantee that the resulting curve has the correct shape.

**What is the symmetry of the graph y = x?**
It is symmetric about the line y = x.

**How can the graph of y = -x be obtained from the graph of y = x?**
By reflecting the graph of y = x about the x-axis.

**Is the graph of y = x symmetric about the y-axis?**
Yes, it is symmetric about the y-axis.

**Can symmetry reduce the computation required for graphing?**
Yes, by utilizing symmetries, computations for one half of the graph can be used to obtain the other half.

**Is the graph of $2x + 2y = 4$ symmetric about the y-axis?**
No, the graph is not symmetric about the y-axis.

**How can the graph of $2x - y = 0$ be obtained from the graph of $y = x$?**
By reflecting the graph of $y = x$ about the x-axis.

**What is the shape of the graph of $y = x$?**
It is a straight line passing through the origin.

**Can the graph of an equation have multiple symmetries?**
Yes, it is possible for a graph to have multiple types of symmetry.

**How can symmetries be useful in mathematical arguments and applied fields?**
Symmetry plays an important role in understanding the structure of the universe and has practical applications in various fields of mathematics and engineering.

# Lecture # 04

**What is the definition of slope?**
The slope is the ratio of the vertical distance to the horizontal distance between two points on a line.

**Does the slope definition apply to vertical lines?**
No, the slope is undefined for vertical lines.

**Can any two distinct points on a non-vertical line be used to calculate its slope?**
Yes, the slope remains the same regardless of which pair of distinct points on the line is used.

**What does a positive slope indicate?**
A positive slope means that the line is inclined upward to the right.

**What does a negative slope indicate?**
A negative slope means that the line is inclined downward to the right.
**What does a zero slope indicate?**
A zero slope means that the line is horizontal.

**How is the angle of inclination related to the slope of a line?**
The slope and angle of inclination are related by the equation $m = \tan(\theta)$, where $\theta$ is the angle measured counterclockwise from the positive x-axis.

**How can you determine if two lines are parallel or perpendicular?**
Two lines are parallel if their slopes are equal ($m1 = m2$), and they are perpendicular if the product of their slopes is -1 ($m1 * m2 = -1$).

**What is the equation of a vertical line?**

The equation of a vertical line passing through (a, 0) is x = a.

**What is the equation of a line passing through a point (x1, y1) with slope m?**
The equation of the line is given by y - y1 = m(x - x1), known as the point-slope form.

**What is the slope-intercept form of a line?**
The slope-intercept form of a line is y = mx + b, where m is the slope and b is the y-intercept.

**What is a first-degree equation in x and y?**
A first-degree equation in x and y is of the form Ax + By + C = 0, where A, B, and C are constants.

**Why is slope important?**
Slope helps us measure steepness, describe changes, and is essential in various applications, such as studying motion and determining relationships between quantities.

**What is the relationship between Fahrenheit and Celsius temperatures?**
Fahrenheit temperature (F) and Celsius temperature (C) are related by a linear equation: F = (9/5)C + 32.

# Lecture # 05

**What will we derive a formula for in this lecture?**
The distance between two points in a coordinate plane.

**What will we study using the formula?**
Equations and graphs of circles.

**How is the distance between two points on a coordinate line calculated?**
By taking the absolute value of the difference between their coordinates.

**What is the formula for finding the distance between two points in a plane?**
sqrt((x2 - x1)^2 + (y2 - y1)^2)

**What theorem is used to derive the distance formula?**
The Pythagorean Theorem.

**How can the distance formula be simplified?**
By taking the absolute value of the differences individually.

**How do we label the points for the distance formula?**

It doesn't matter which point is labeled as $(x1, y1)$ or $(x2, y2)$.

**How can we determine if three points form a right triangle?**
By checking if the sum of the squares of the two shorter sides equals the square of the longest side (Pythagorean theorem).

**What is the midpoint formula?**
The coordinates of the midpoint between two points are the average of their coordinates.

**How is the midpoint formula derived?**
By considering similar triangles formed by the midpoint and the two points on the x-axis and y-axis.

**What is the equation of a circle in standard form?**
$(x - x0)^2 + (y - y0)^2 = r^2$

**How can the equation of a circle be written in expanded form?**
$Ax^2 + Ay^2 + Dx + Ey + F = 0$

**What is the special importance of the unit circle?**
It is a circle centered at the origin with a radius of 1.

**What are the three cases for the equation of a circle?**
Circle, point, or no graph (degenerate cases).

**What is the shape of the graph for a quadratic equation in x?**
A parabola.

**What is the line of symmetry for a parabola?**
A vertical line parallel to the y-axis.

**What is the vertex of a parabola?**
The highest or lowest point on the curve.

**How can we solve an inequality graphically?**
By determining the values of x that satisfy the inequality by analyzing the graph.

**How can we find the x-intercepts of a quadratic equation?**
By setting $y = 0$ and solving for x using the quadratic formula.

**What is the distance formula for a ball thrown straight up?**
$s = -4.9t^2 + 24.5t$